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Rectified Language Flow

Straight-Path Flow Matching in Token Embedding Space Beats Autoregressive Baselines at 15–19× Inference Speedup

By Baiyu Su, Runlong Liao, Lizhang Chen, Qiang Liu

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Rectified Language Flow (ReLF) generates text by training a flow model to move noise continuously toward token embeddings along straight (rectified) paths, enabling accurate generation in as few as one to four function evaluations. At batch size 1 it produces 1,024-token sequences 15–19× faster than a matched autoregressive baseline, while achieving lower generative perplexity on OpenWebText and outperforming all prior continuous-flow language models.

AT A GLANCE

Problem: Autoregressive models are slow at inference due to serial token generation; discrete diffusion models require many costly denoising steps over a large discrete space.

Why it matters: Fast, high-quality text generation is critical for interactive AI, serving efficiency, and real-time applications such as code completion and dialogue.

Method: Continuous flow matching in Hellinger-SVD token embeddings with self-conditioning guidance, regularized reflow for distillation, and singular logits for discrete commitment.

Result: Lower perplexity than autoregressive baselines on OpenWebText; 15–19× faster generation at batch size 1 with 4 function evaluations.

Evidence: OpenWebText language modeling benchmarks across three model sizes; ablation over number of function evaluations.

The Big Picture

Large language models have converged on autoregressive (AR) generation: at each step the model conditions on all previous tokens and samples the next one. This left-to-right serial process is the primary bottleneck at inference time—each token requires a full forward pass, and a sequence of T tokens requires T passes.

Diffusion language models offer an alternative: start from a noisy sequence and iteratively refine it over K denoising steps. Because denoising steps can be parallelized across positions, the number of forward passes is $K \ll T$ for long sequences. But discrete diffusion models operate over a large categorical space at each position, introducing its own inefficiencies, and K is typically still in the tens to hundreds.

ReLF takes a third approach: flow matching in a continuous embedding space. A flow model is trained to transport a simple noise distribution p_0 (e.g., Gaussian) to the distribution of token embeddings p_1. The rectified flow objective enforces straight paths between noise and data, enabling accurate generation with as few as 1–4 function evaluations.

Why Existing Approaches Fall Short

Autoregressive models require T serial forward passes for a length-T sequence. Speculative decoding and other parallelization heuristics reduce this in practice but don't change the fundamental asymptotic cost.

Discrete diffusion models work directly in token space, avoiding the need to commit to embeddings. However, the discrete state space is large (vocabulary $\sim$ 50k), and the denoising objective must define meaningful noise over discrete categories. Prior models report 32–64 steps to achieve quality competitive with AR models.

Prior continuous-flow language models (ELF, DiLaDiff) use learned continuous embeddings that are unstable to train and require reconstruction losses that decouple the flow from the discrete output space. Committing a continuous embedding to a discrete token at generation time via nearest-neighbor lookup introduces discretization error that accumulates over the sequence.

Core Mechanism

ReLF's four key components work together to address these issues:

1. Hellinger-SVD Embeddings. Token embeddings are initialized from the SVD of the Hellinger-transformed co-occurrence matrix of the training corpus. This produces a geometry in which frequent co-occurring tokens are nearby, providing a meaningful flow target that is fixed throughout training. Crucially, no embedding training is needed.

2. Self-Conditioning Guidance. At each training step the model is first run with the corrupted input x_t to produce a preliminary estimate $\hat{x}_0$. On the same forward pass (with gradients detached), $\hat{x}_0$ is concatenated to x_t and passed to the model again. A cross-entropy loss between $\hat{x}_0$ and the true x_0 trains the guidance branch. This conditioning helps the model resolve ambiguous intermediate states near $t = 0.5$.

3. Regularized Reflow. After initial training, a reflow stage distills the learned curved flow into a straighter one by training on (x_0, x_1) pairs sampled from the original model's trajectories. A regularization term penalizes degenerate straight lines that pass outside the data manifold, preventing the reflow from creating diversity collapse.

4. Singular Logits. As $t \to 1$ the flow approaches the data manifold and must commit to a discrete token. Singular logits is a differentiable mechanism that computes token logits from the flow's velocity direction at $t = 1$, rather than from the embedding value itself, enabling sharp, gradient-compatible token selection.

Technical Formulation

Let $x_0 \sim \mathcal{N}(0, I)$ be noise and $x_1 \sim p_{\text{data}}$ be a token embedding sequence. The rectified flow objective minimizes:

$$\mathcal{L}_{\text{flow}} = \mathbb{E}_{t,x_0,x_1}\left[\|v_\theta(x_t, t) - (x_1 - x_0)\|^2\right]$$

where $x_t = (1-t)x_0 + t\,x_1$ and v_θ is the learned velocity field.

The self-conditioning guidance loss augments training with:

$$\mathcal{L}_{\text{sc}} = -\log p_\theta(x_1 \mid \hat{x}_0, x_t, t)$$

where $\hat{x}_0 = x_t + (1-t)\,v_\theta(x_t, t)$ is the preliminary clean estimate.

The regularized reflow loss is:

$$\mathcal{L}_{\text{reflow}} = \mathcal{L}_{\text{flow}}^{\text{straight}} + \lambda \cdot \mathbb{E}\big[\text{dist}(x_t^{\text{reflow}}, \mathcal{M})\big]$$

where $\mathcal{M}$ is an approximation of the data manifold and λ is a regularization weight. Singular logits are defined as:

$$\ell_v = W_{\text{vocab}}\, v_\theta(x_1, 1) \cdot \|W_{\text{vocab}}\|^{-1}$$

mapping the terminal velocity to logits via the vocabulary matrix.

Learning or Inference Procedure

Training:

1. Pre-compute Hellinger-SVD embeddings from corpus statistics.
2. Train v_θ with $\mathcal{L}_{\text{flow}} + \alpha \cdot \mathcal{L}_{\text{sc}}$ for N steps.
3. Collect (x_0, x_1) pairs from teacher model trajectories.
4. Fine-tune with $\mathcal{L}_{\text{reflow}}$ for $M \ll N$ additional steps.

Inference: Sample $x_0 \sim \mathcal{N}(0, I)$, integrate the velocity field over $K \in \{1, 2, 4, 8, 16, 64\}$ Euler steps, apply singular logits at $t = 1$ to decode token distributions, and sample.

What the Guarantee Says

The rectified flow objective guarantees that the expected trajectory length $\mathbb{E}[\int_0^1 \|v_\theta\|\,dt]$ is minimized when trajectories are straight, which corresponds to the lower bound on the number of function evaluations required for a given discretization error. While this is an asymptotic argument, the regularized reflow empirically achieves near-straight paths and the 1-step sample quality confirms the practical benefit. No PAC-style generalization bound is provided; the guarantee is about transport efficiency, not sample quality.

Experimental Findings

- **Perplexity:** On OpenWebText, with 64 function evaluations, ReLF achieves **lower perplexity than AR baselines** at matched parameter counts for all three model sizes (90M, 350M, 1.2B).
- **Speed:** At 4 function evaluations, ReLF generates 1,024-token sequences **15–19× faster** than the AR baseline at batch size 1, with less than 0.5 perplexity units of degradation relative to the 64-step setting.
- **Comparison to ELF:** ReLF outperforms ELF at all model sizes by 1.2–2.8 perplexity points.
- **Diversity:** At matched perplexity, ReLF self-BLEU scores are lower (more diverse) than the AR baseline, suggesting the flow's sampling process does not collapse to near-deterministic outputs.

Ablations and Interpretation

Hellinger-SVD vs. learned embeddings: Learned embeddings reduce final perplexity by 0.3 points on average but add training instability; the fixed Hellinger-SVD embeddings are more robust and are used in all reported experiments.

Without self-conditioning: Removing self-conditioning increases perplexity by 1.8–3.4 points, with the largest degradation at small K (few steps). This confirms that guidance is most critical in the low-step regime.

Without reflow: Without the reflow stage, 4-step generation degrades by 5.1 perplexity points relative to the 64-step setting. With reflow, the gap closes to < 0.5 points.

Singular logits vs. nearest neighbor: Replacing singular logits with nearest-neighbor lookup in embedding space increases perplexity by 0.7–1.1 points and degrades diversity. The directional signal at $t = 1$ is more discriminative than the position signal.

Reference

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